Code No: 121AL **R15**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech I Year Examinations, July - 2021
MATHEMATICAL METHODS
(Common to EEE, ECE, CSE, EIE, IT, ETM)

Time: 3 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Interpolate the value of y at $x = 2.5$ from the following data.

x:	0	1	2	3
y:	2	5	13	35

b) Prove that $\Delta\nabla = \Delta - \nabla$ and $\nabla = E^{-1}\Delta$, where Δ, ∇ and E are the forward, backward and shifting operators respectively. [7+8]

2. Fit a straight line of the form $y = a + bx$ and a quadratic curve of the form $y = a + bx + cx^2$ to the data given below. [15]

x:	1	2	3	4
y:	10	5	2	1

3. Solve the equations $10x + y + z = 12, 2x + 10y + z = 13, 2x + 2y + 10z = 104$ by Gauss-Seidel iteration method. [15]

4. Apply Runge-Kutta method of order 4, solve $\frac{dy}{dx} = \frac{y^2 - x^2}{y^2 + x^2}$, $y(0) = 1$ *at* $x = 0.1, 0.2$. [15]

5. Find the Fourier series expansion of $f(x) = \begin{cases} \pi + x, & -\pi < x \le 0 \\ 0, & 0 < x < \pi \end{cases}$, $f(x + 2\pi) = f(x)$ and hence show that $1 + \frac{1}{3^2} + \frac{1}{5^2} + \ldots\ldots = \frac{\pi^2}{8}$. [15]

6.a) Solve $x(y - z)p + y(z - x)q = z(x - y)$.

b) Find a partial differential equation by eliminating the arbitrary functions f and g from $z = f(x)\, g(y)$. [7+8]

7.a) If $\vec{F} = (x + y + 1)\hat{i} - \hat{j} - (x + y)\hat{k}$, show that $\vec{F} \cdot Curl\ \vec{F} = 0$.

b) Show that $\nabla^2\left(r^n\right) = n(n+1)r^{n-2}$. [7+8]

8. Verify Gauss's divergence theorem for $\vec{F} = x^2\hat{i} + z\hat{j} + yz\hat{k}$ taken over the cube bounded by $x = 0, x = 1, y = 0, y = 1, z = 0$ *and* $z = 1$. [15]

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